

Project on Local Ingress controller

#Below is the repository.

ingress-backend-repo

ingress-fronten-repo

```
alam@DESKTOP-VK19M09:project$ git clone
https://github.com/Avejlanjekar/Employee_management_system.git
Cloning into 'Employee_management_system'...
```

```
alam@DESKTOP-VK19M09:backend$ docker build -t ingress-backend .
0.0s
```

=> => unpacking to docker.io/library/ingress-backend:latest

```
alam@DESKTOP-VK19M09:backend$ docker tag ingress-backend:latest qamre/ingress-backend-
repo:v1
```

```
alam@DESKTOP-VK19M09:backend$ docker push qamre/ingress-backend-repo:v1
```

The push refers to repository [docker.io/qamre/ingress-backend-repo]

```
alam@DESKTOP-VK19M09:frontend$ docker tag ingress-frontend:latest qamre/ingress-fronten-
repo:v1
```

```
alam@DESKTOP-VK19M09:frontend$ docker push qamre/ingress-fronten-repo:v1
```

=====

```
alam@DESKTOP-VK19M09:project$ cat frontend-deployment.yaml
```

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
  name: frontend
```

```
spec:
```

```
  replicas: 2
```

```
selector:
  matchLabels:
    app: frontend
template:
  metadata:
    labels:
      app: frontend
  spec:
    containers:
      - name: frontend
        image: qamre/ingress-fronten-repo:v1
        ports:
          - containerPort: 80
```

lam@DESKTOP-VK19M09:project\$ cat frontend-service.yaml

```
apiVersion: v1
kind: Service
metadata:
  name: frontend-service
spec:
  type: ClusterIP
  selector:
    app: frontend
  ports:
    - port: 80
      targetPort: 80
```

alam@DESKTOP-VK19M09:project\$ cat backend-deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
```

```
metadata:
  name: backend
spec:
  replicas: 2
  selector:
    matchLabels:
      app: backend
  template:
    metadata:
      labels:
        app: backend
    spec:
      containers:
        - name: backend
          image: qamre/ingress-backend-repo:v2
          ports:
            - containerPort: 5000
          env:
            - name: DB_HOST
              value: "mysql"
            - name: DB_USER
              value: "root"
            - name: DB_PASSWORD
              value: "rootpassword"
            - name: DB_NAME
              value: "employees"
            - name: DB_PORT
              value: "3306"
```

alam@DESKTOP-VK19M09:project\$ cat backend-service.yaml

apiVersion: v1

```
kind: Service
metadata:
  name: backend
spec:
  type: ClusterIP
  selector:
    app: backend
  ports:
    - port: 5000
      targetPort: 5000
```

```
cat ingress.yaml
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: myapp-ingress
spec:
  ingressClassName: nginx
  rules:
    - host: myapp.local
      http:
        paths:
          - path: /api
            pathType: Prefix
            backend:
              service:
                name: backend
                port:
                  number: 5000
          - path: /
```

```
pathType: Prefix
backend:
  service:
    name: frontend-service
  port:
    number: 80
```

```
alam@DESKTOP-VK19M09:project$ cat mysql.yaml
```

```
apiVersion: v1
kind: Service
metadata:
  name: mysql
spec:
  clusterIP: None
  selector:
    app: mysql
  ports:
    - port: 3306
      targetPort: 3306
---
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: mysql
spec:
  serviceName: mysql
  replicas: 1
  selector:
    matchLabels:
      app: mysql
```

```
template:
  metadata:
    labels:
      app: mysql
  spec:
    containers:
      - name: mysql
        image: mysql:8.0
        ports:
          - containerPort: 3306
        env:
          - name: MYSQL_ROOT_PASSWORD
            value: "rootpassword"
          - name: MYSQL_DATABASE
            value: "employees"
        volumeMounts:
          - name: mysql-storage
            mountPath: /var/lib/mysql
    volumeClaimTemplates:
      - metadata:
          name: mysql-storage
        spec:
          accessModes:
            - ReadWriteOnce
          resources:
            requests:
              storage: 5Gi
```

alam@DESKTOP-VK19M09:project\$

lam@DESKTOP-VK19M09:project\$ kubectl get po

NAME	READY	STATUS	RESTARTS	AGE
backend-69bc66bf9b-bckpt	1/1	Running	0	45m
backend-69bc66bf9b-pm5xq	1/1	Running	0	46m
frontend-5f8c64b745-dq2jt	1/1	Running	1 (4h35m ago)	4h58m
frontend-5f8c64b745-pzqr9	1/1	Running	1 (4h35m ago)	4h58m
mysql-0	1/1	Running	0	63m

```
mysql-0          1/1   Running  0          63m
alam@DESKTOP-VK19M09:project$ kubectl get svc
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
backend	ClusterIP	10.98.245.39	<none>	5000/TCP	4h59m
frontend-service	ClusterIP	10.103.100.228	<none>	80/TCP	5h8m
mysql	ClusterIP	None	<none>	3306/TCP	66m

```
alam@DESKTOP-VK19M09:project$ kubectl get ingress
```

NAME	CLASS	HOSTS	ADDRESS	PORTS	AGE
myapp-ingress	nginx	myapp.local	192.168.49.2	80	4h59m

Employee Management

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[Employees](#)
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Employee Management System

Manage your organization's employees efficiently from one place.

View Employees

Employee Management

myapp.local says
Employee added successfully

Home Employees Add Employee

Add Employee

Name

abcd

Email

abcd@a.com

Department

finance

Salary

100

Add Employee

Employees

Manage all employees

+ Add Employee

Total Employees

3

Departments

3

Total Salary

₹2100

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